AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

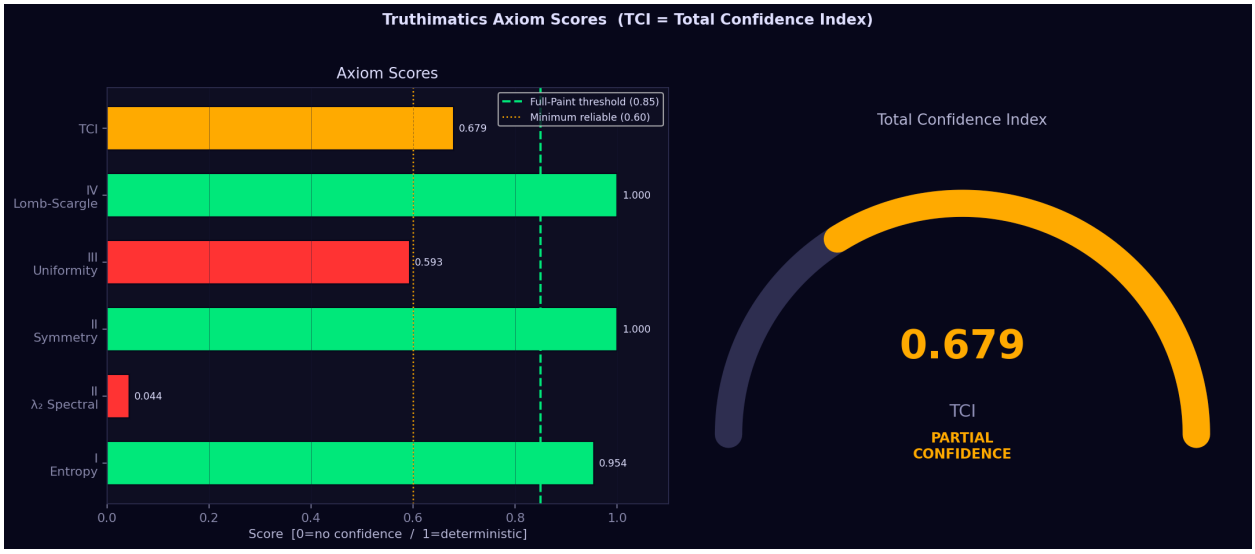
Specimen ID	AXIOM_304F942756
Taxa	Styracosaurus
Clade	Ornithischian
PBDB Records	10
Mass (regression)	2932.0 kg (Campione & Evans 2012)
Mass (volumetric)	2813.7 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	2.962 m³
Validity Score	1.0000 / 1.0000
TCI	0.6795 (partial)
Axiom I Entropy	0.9543
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.5932
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.377, 1.085, 0.0]
Generated	2026-04-30 23:57 UTC

Ratio Violations

• skull/femur=1.923 outside [0.30,1.80]

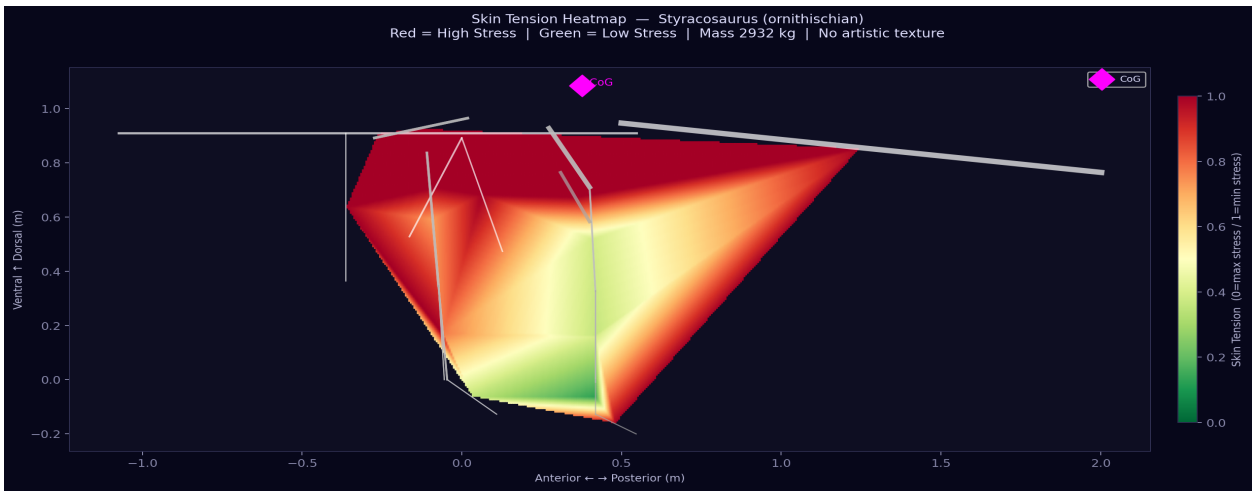
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



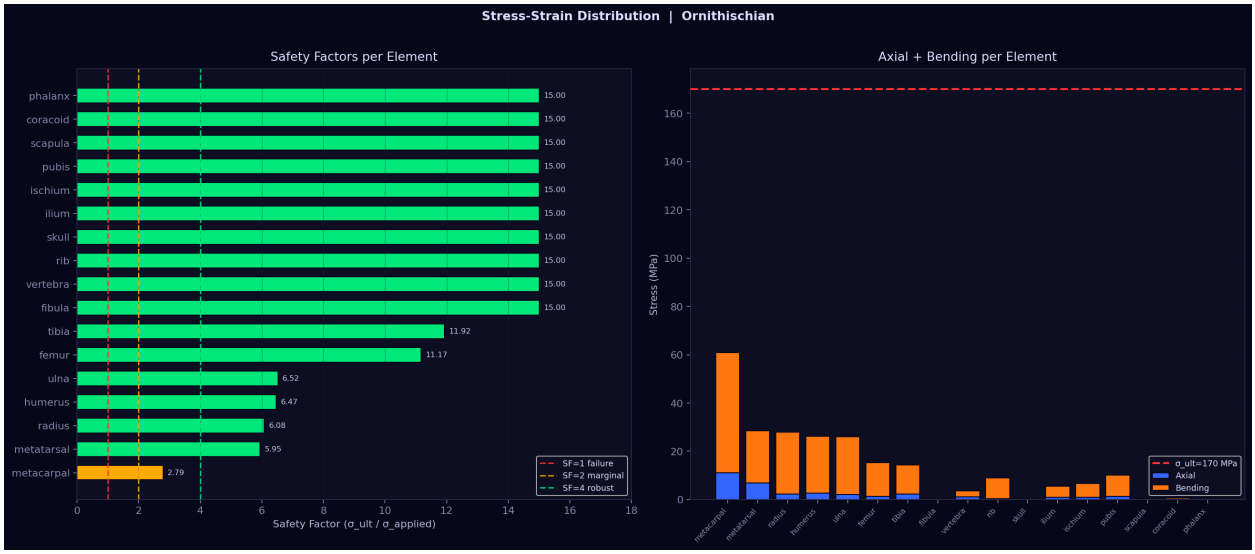
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

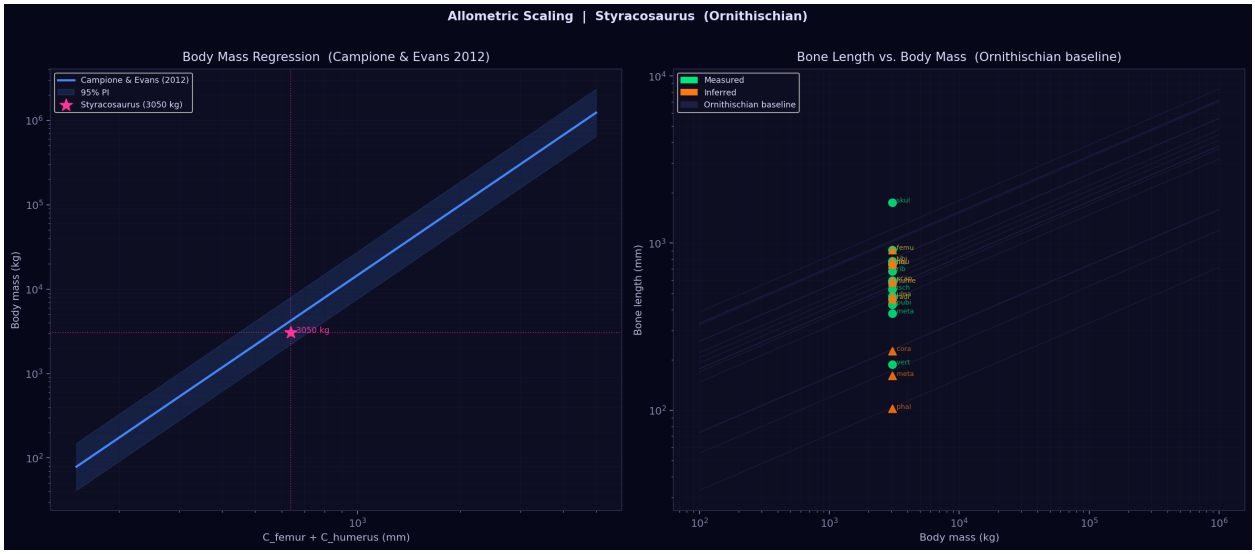


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.243	0.450	0.692	15.000	OK
femur	1.341	13.884	15.225	11.166	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	2.585	23.675	26.260	6.474	OK
ilium	0.853	4.627	5.480	15.000	OK
ischium	0.884	5.827	6.710	15.000	OK
metacarpal	11.125	49.761	60.887	2.792	WARNING
metatarsal	6.907	21.679	28.586	5.947	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	1.334	8.758	10.092	15.000	OK
radius	2.242	25.742	27.984	6.075	OK
rib	0.269	8.674	8.943	15.000	OK
scapula	0.126	0.443	0.568	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	2.206	12.053	14.259	11.922	OK
ulna	2.084	23.981	26.065	6.522	OK
vertebra	1.195	2.413	3.608	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.07 m	Femur + tibia + metatarsal
Estimated Speed	13.4 km/h (3.71 m/s)	Gait: Walk | Fr=0.677
Stride Length	4.225 m	Alexander (1976) / Thulborn (1990)
Bite Force	104716.6 N (10674.5 kgf)	Ornithischian regression (Norman & Weishampel 1991)
Skull Length	1750.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.4074	Circumference / length (dimensionless)
Limb Strength Index	4.3e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	8.22 m	Allometric from femur length
Est. Neck Length	0.9 m	Allometric from femur length
Est. Tail Length	2.3 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	226.7	164.1	2446.09	3053.4	Constraint	0.40	15.00
femur	910.0	118.0	5075.34	6335.6	Measured	0.88	11.17
femur_right	910.0	118.0	5075.34	6335.6	Symmetry	0.84	11.17
fibula	752.0	38.0	307.03	322.0	Measured	0.88	15.00
fibula_right	752.0	38.0	307.03	322.0	Symmetry	0.84	15.00
humerus	580.0	85.0	1678.52	2095.3	Measured	0.88	6.47
humerus_right	580.0	85.0	1678.52	2095.3	Symmetry	0.84	6.47
ilium	748.0	148.0	6562.74	8192.3	Measured	0.88	15.00
ilium_right	748.0	148.0	6562.74	8192.3	Symmetry	0.84	15.00
ischium	528.0	86.0	1564.19	1952.6	Measured	0.88	15.00
metacarpal	160.8	39.7	37.83	31.1	Constraint	0.65	2.79
metatarsal	380.0	52.0	411.58	513.8	Measured	0.88	5.95
phalanx	102.0	32.8	13.26	10.3	Constraint	0.40	15.00
pubis	428.0	70.0	840.04	1048.6	Measured	0.88	15.00
radius	462.0	54.0	539.62	673.6	Measured	0.88	6.08
radius_right	462.0	54.0	539.62	673.6	Symmetry	0.84	6.08
rib	680.0	34.0	314.87	393.0	Measured	0.88	15.00
scapula	598.0	228.0	12451.77	15543.5	Measured	0.88	15.00
scapula_right	598.0	228.0	12451.77	15543.5	Symmetry	0.84	15.00
skull	1750.0	720.0	363381.74	453609.4	Measured	0.88	15.00
tibia	780.0	92.0	2644.42	3301.0	Measured	0.88	11.92
tibia_right	780.0	92.0	2644.42	3301.0	Symmetry	0.84	11.92
ulna	480.0	56.0	602.94	752.7	Measured	0.88	6.52
ulna_right	480.0	56.0	602.94	752.7	Symmetry	0.84	6.52
vertebra	188.0	125.0	1176.62	1468.8	Measured	0.88	15.00